MACS+ - Membrane Action Of Composite Structures In Case Of Fire

R E P O R T

1. Project properties

Project name:	Atlantic Park Phase 2 Unit 7	Prepared by:	Kevin Kurniawan
Client name:		Company name:	Fire Dynamics Group Limited
Client company:	Winvic Construction Limited	Date:	29 April 2026, 06:47
Job number:	0563	Checked by:	
Comments:	Second Floors Slab - Plant Deck		

2. General arrangement

• Spans

Span 1: 7.3 m

Span 2: 7.48 m

• Unprotected beams

Number of internal unprotected beams: 2

3. Deck details

• Deck properties

Deck:	Multideck 60	Type:	Trapezoidal
Depth:	60 mm	Top flange:	142 mm
Pitch:	323 mm	Bottom flange:	119 mm
Stiffener height:	9 mm		

4. Slab details

• Concrete

Concrete type:	Normal	Slab depth:	150 mm
		Cylinder compressive strength of concrete (f_{ck}):	30 N/mm²

• Mesh

Mesh type:	A193		
Longitudinal mesh area:	193 mm²/m	Bar size:	7 mm
Transverse mesh area:	193 mm²/m	Bar size:	7 mm
Average mesh axis distance:	52 mm	Mesh yield stress:	500 N/mm²

5. Beams details

• Unprotected beams

Beam type:	Solid beam		
Section family:	UK sections	Steel grade:	S355
		Degree of shear connection:	80 %
Section size:	UB 356x127x33		
Details:	h = 349 mm, b = 125.4 mm, t_w = 6 mm, t_f = 8.5 mm		

• Side A perimeter beam

Beam type:	Solid beam		
Section family:	UK sections	Steel grade:	S355
Section size:	UB 356x127x33		
Details:	h = 349 mm, b = 125.4 mm, t_w = 6 mm, t_f = 8.5 mm		
Beam location:	Internal beam	Construction type:	Composite
		Degree of shear connection:	80 %

• Side B perimeter beam

Beam type:	Solid beam		
Section family:	UK sections	Steel grade:	S355
Section size:	UB 457x152x52		
Details:	h = 449.8 mm, b = 152.4 mm, t_w = 7.6 mm, t_f = 10.9 mm		
Beam location:	Edge beam	Construction type:	Composite
		Degree of shear connection:	80 %

• Side C perimeter beam

Beam type:	Solid beam		
Section family:	UK sections	Steel grade:	S355
Section size:	UB 356x127x33		
Details:	h = 349 mm, b = 125.4 mm, t_w = 6 mm, t_f = 8.5 mm		
Beam location:	Internal beam	Construction type:	Composite
		Degree of shear connection:	80 %

• Side D perimeter beam

Beam type:	Solid beam		
Section family:	UK sections	Steel grade:	S355
Section size:	UB 457x152x52		
Details:	h = 449.8 mm, b = 152.4 mm, t_w = 7.6 mm, t_f = 10.9 mm		
Beam location:	Edge beam	Construction type:	Composite
		Degree of shear connection:	80 %

6. Loading details

• Normal (Cold)

Leading variable action:	7.5 kN/m²
Accompanying variable action:	0 kN/m²
Dead load including beam, excluding slab:	0.5 kN/m²
Calculated slab weight including mesh:	2.83 kN/m²

• Fire (Hot)

Combination factor for permanent action:	1.0
Combination factor for leading variable action:	0.5
Combination factor for other variable action:	0.3

7. Fire & Analysis

• Parametric temperature-time curve

Compartment length:	37.1 m	Window height:	3.4 m
Compartment width:	7.26 m	Window length:	35.9 m
Compartment height:	3.4 m	Glazing breakage:	87.254 %
Fire load:	503.50 MJ/m²	Wall lining (B) factor:	1700 J/m²s½K
Growth rate:	Medium	Combustion factor:	0.8

8. Summary output

• Default mesh direction

Maximum unity factor:	0.42	**Floor slab adequate**
Factored load in fire:	7.08 kN/m²	
Fire curve:	Parametric temperature-time curve	

9. Default mesh direction

• Tabular results

Longitudinal mesh area:	193 mm²/m	Bar size:	7 mm
Transverse mesh area:	193 mm²/m	Bar size:	7 mm

Factored load in fire: 7.08 kN/m²

Time	Beam	Mesh	Slab top	Slab bottom	Beam capacity	Maximum allowable deflection	Slab yield	Enhance-ment	Slab capacity	Total capacity	Unity factor
mins	°C	°C	°C	°C	kN/m²	mm	kN/m²		kN/m²	kN/m²	
0	20	20	20	20	27.22	158	2.13	1.75	3.73	30.94	0.23
4	64	21	20	75	27.22	174	2.13	1.83	3.89	31.11	0.23
8	188	28	20	132	27.22	190	2.13	1.91	4.06	31.28	0.23
12	357	39	21	241	27.22	220	2.13	2.06	4.38	31.60	0.22
16	513	50	22	345	20.39	249	2.13	2.20	4.69	25.08	0.28
20	618	69	24	430	11.92	273	2.13	2.32	4.93	16.86	0.42
24	541	75	27	342	18.04	247	2.13	2.19	4.66	22.71	0.31
28	390	78	30	224	27.22	213	2.13	2.02	4.30	31.52	0.22
32	276	80	33	161	27.22	194	2.13	1.93	4.11	31.32	0.23
36	206	82	37	129	27.22	184	2.13	1.88	4.00	31.22	0.23
40	157	83	40	109	27.22	177	2.13	1.85	3.93	31.15	0.23
44	123	84	43	94	27.22	172	2.13	1.82	3.88	31.10	0.23

Maximum unity factor: 0.42 **Floor slab adequate**

- Perimeter beam check

Side A	Beam type:	Solid beam	Composite	Internal beam
	Section size:	UB 356x127x33		
	Required moment resistance in fire situation:	0 kNm		
	Line load in fire situation:	0 kN/m		
	Shear connection:	80 %		
	Degree of utilization:	0.45		
	Critical temperature:	628 °C		
Side B	Beam type:	Solid beam	Composite	Edge beam
	Section size:	UB 457x152x52		
	Required moment resistance in fire situation:	0 kNm		
	Line load in fire situation:	0 kN/m		
	Shear connection:	80 %		
	Degree of utilization:	0.45		
	Critical temperature:	631 °C		
Side C	Beam type:	Solid beam	Composite	Internal beam
	Section size:	UB 356x127x33		
	Required moment resistance in fire situation:	0 kNm		
	Line load in fire situation:	0 kN/m		
	Shear connection:	80 %		
	Degree of utilization:	0.45		
	Critical temperature:	628 °C		
Side D	Beam type:	Solid beam	Composite	Edge beam
	Section size:	UB 457x152x52		
	Required moment resistance in fire situation:	0 kNm		
	Line load in fire situation:	0 kN/m		
	Shear connection:	80 %		
	Degree of utilization:	0.45		
	Critical temperature:	631 °C		

10. Graphical output